Plane Wave Propagation in Lossy Media

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Lecture Objective

By the end of this lecture, you will be prepared to:

- **Derive and examine the behavior of a plane wave propagating in a lossy medium.**

Helmholtz Equation for Electric Field

- From previous lectures

$$\nabla^2 \tilde{\vec{E}} - \gamma^2 \tilde{\vec{E}} = 0$$

$$\tilde{\vec{H}} = \frac{1}{\eta} \hat{k} \times \tilde{\vec{E}}$$

Lossy Medium

- A lossy medium is one for which at least one of the conditions listed below applies:

$$\varepsilon'' > 0$$

$$\mu'' > 0$$

$$\sigma > 0$$

Question 1

- What are the expressions for propagation constant and intrinsic impedance in terms of impedivity and admittivity of the medium?

Propagation, Attenuation and Phase Constants

$$\gamma = \sqrt{yz} = \sqrt{(\sigma + j\omega\varepsilon)j\omega\mu}$$
$$= \sqrt{(\sigma + \omega\varepsilon'' + j\omega\varepsilon')(\omega\mu'' + j\omega\mu')}$$
$$= \alpha + j\beta$$

Intrinsic Impedance

$$\eta = \sqrt{\frac{z}{y}} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$= \sqrt{\frac{\omega\mu'' + j\omega\mu'}{\sigma + \omega\varepsilon'' + j\omega\varepsilon'}} \quad = |\eta| e^{j\theta_\eta}$$

The implication of complex intrinsic impedance is that the electric field intensity and magnetic field intensity of the plane wave are no longer in phase.

Plane Wave Solution in Frequency-Domain Continued

- Let's assume a time-harmonic plane wave traveling in the +z-direction with only an x-component of the electric field.

$$\tilde{\vec{E}} = \hat{x}\tilde{E}_x(z)$$

$$\frac{\partial}{\partial x} \equiv 0$$

$$\frac{\partial}{\partial y} \equiv 0$$

Uniform Plane Wave Solution

- Helmholtz equation for electric field becomes

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} - \gamma^2 \tilde{E}_x = 0$$

Plane Wave Solution in Frequency-Domain

- Solutions to the above DE can be written as

$$\tilde{E}_x(z) = E_{x0} e^{\mp \gamma z} = E_{x0} e^{\mp \alpha z} e^{\mp j\beta z}$$

- These solutions are time-harmonic waves:

$$E_x(z,t) = |E_{x0}|\, \mathrm{e}^{\mp \alpha z} \cos(\omega t \mp \beta z + \langle E_{x0})$$

$$H_y(z,t) = \pm \frac{|E_{x0}|}{|\eta|} \mathrm{e}^{\mp \alpha z} \cos\left(\omega t \mp \beta z + \theta_\eta + \langle E_{x0}\right)$$

Question 2

- Reduce the equations for electric and magnetic field intensities on the previous slide to the expressions for a lossless medium from a previous lecture.

Phase Velocity

- The wavefronts (constant phase surfaces) are defined by $z = \text{constant}$.
- The velocity of a wavefront is given by differentiating the equation for a constant wavefront w.r.t. t and solving for dz/dt:

$$\omega t \mp \beta z = \text{constant}$$

$$v_p = \frac{dz}{dt} = \pm \frac{\omega}{\beta}$$

Skin Depth

- The **skin depth** (a.k.a. **depth of penetration**) denoted by δ_s is the distance over which the amplitude of a traveling plane wave decreases by a factor of e^{-1}.
- The skin depth tells us how deep an electromagnetic wave of specified frequency can penetrate into lossy medium.

$$\delta_s = \frac{1}{\alpha}$$

- The skin depth is analogous to the time constant in a RC or RL circuit.

$$e^{-\frac{t}{\tau}} \Leftrightarrow e^{-\frac{z}{\delta_s}}$$

Summary

- The propagation constant of a plane wave propagating in a lossy medium is a complex number; the intrinsic impedance is as well.
- The amplitude of the electromagnetic wave propagating in the medium decays exponentially.
- Skin depth tells us the depth of penetration of an electromagnetic wave into a lossy medium; the quantity is often used when describing good conductors.

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